

contracts serve the same purpose as legal contracts in the real world — one can query them and add a state to the blockchain.

A lot of users interact with the smart contract at the same time. Therefore, it should be written in a language that is non-blocking and asynchronous.

Some of the most popular languages used for writing contracts are listed below.

Solidity: This is an object-oriented language used to write smart contracts. It is a lot like JavaScript, which makes it easy to understand. You can find more details about it at <https://solidity.readthedocs.io/en/v0.5.4/style-guide.html>.



Figure 1: Solidity – A contract oriented high-level language

Go: Golang, a language developed by Google, is used in Hyperledger to write chaincodes. It's a language that takes advantage of modern hardware architecture. It also has native garbage collection that removes the need for developers to do their own memory management. Further details can be found in the Golang home page at <https://golang.org/>.

NodeJS: This is an asynchronous language, which is used for application code as well as for smart contracts.

After figuring out the language, we need a development environment. Development environments come with a lot of configuration that will help throughout the process of application development.



Figure 2: NodeJS - Non-blocking and event driven

Integrated development environments (IDE)

Often, we need to develop a proof-of-concept (POC) before starting with the actual application. Here, let's concentrate only on smart contracts and leave out the other development aspects like testing, debugging, integration, etc, that are handled by IDEs.

Some popular open source IDEs are listed below.

Remix: This open source tool can be accessed from a browser or installed in a local system. It is written in JavaScript. It can be used to write, deploy and then run smart contracts for the Ethereum network.

Features

- It provides features like testing, debugging and deploying of smart contracts.
- It also calculates the gas to be used for carrying out a transaction. Gas is a unit that measures the computational effort required to carry out a certain operation. You can find out more about gas at <https://blockgeeks.com/guides/thereum-gas-step-by-step-guide/>. Figure 3 shows its features such as analysis, compilation, testing and debugging.

TronStudio: This is a comprehensive IDE that runs the code for TRON Virtual Machine (TVM). It

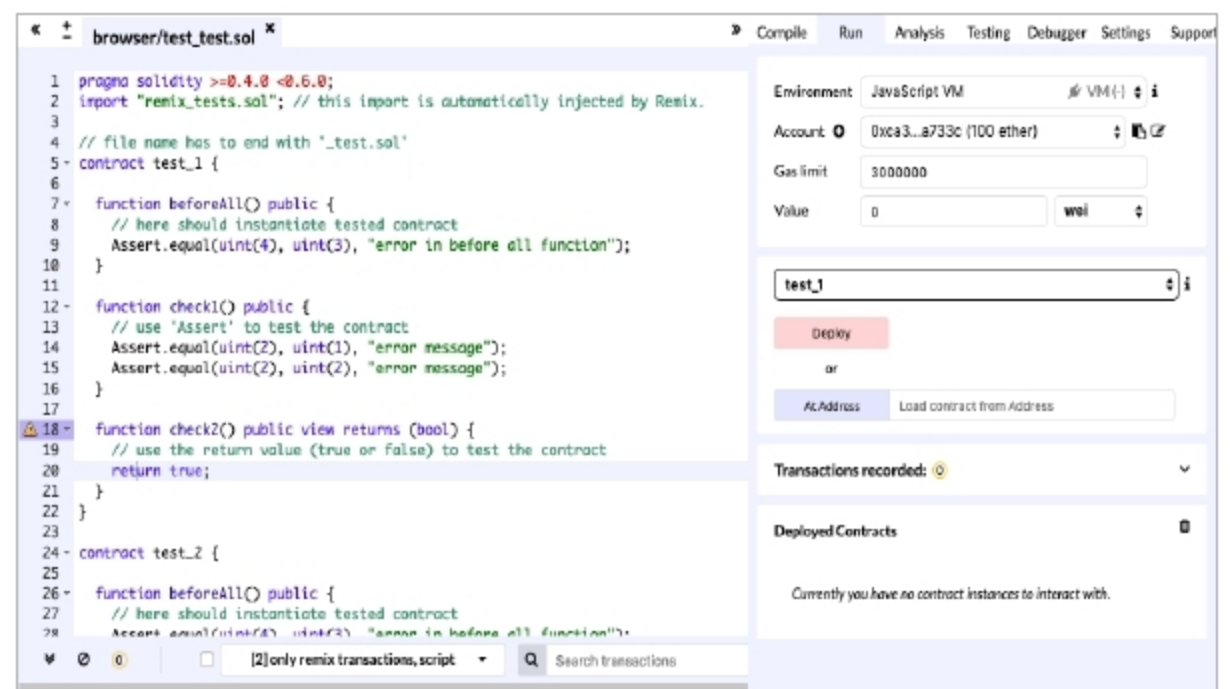


Figure 3: Remix browser

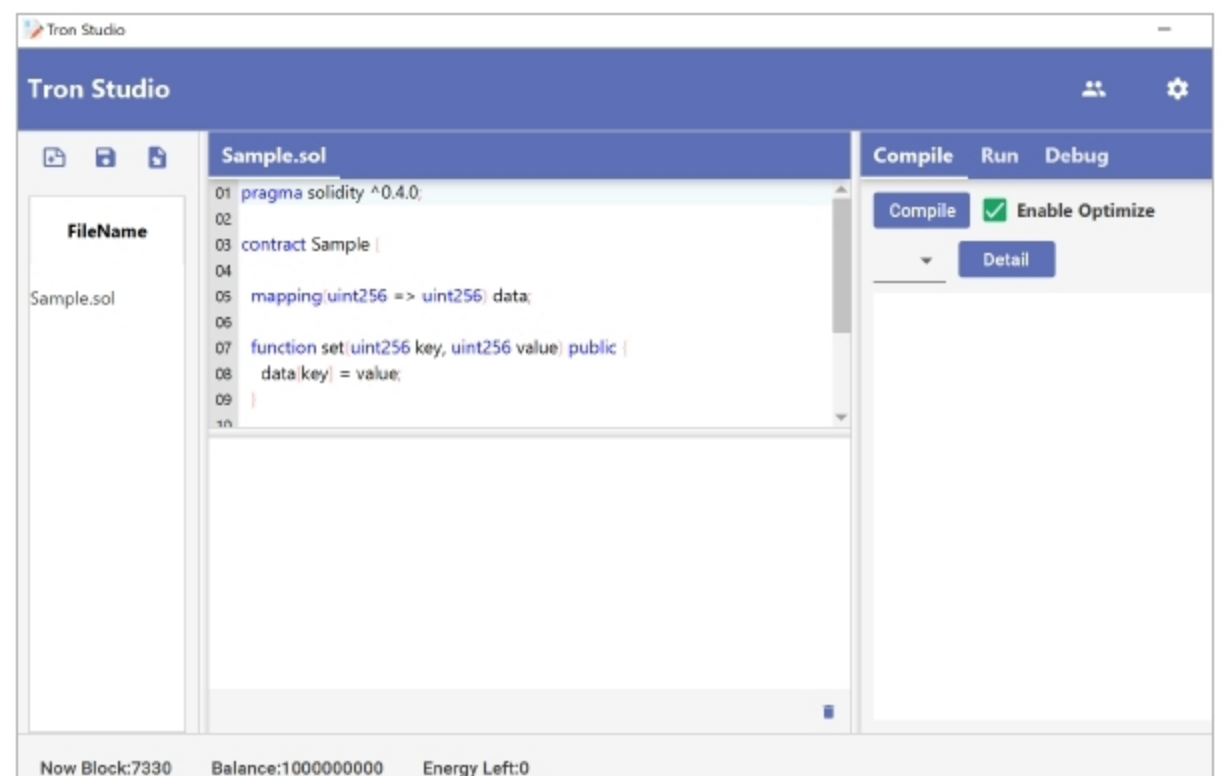


Figure 4: TronStudio